

Sentiment-to-Price Correlation Analysis Report

1. Executive Summary

This report analyzes the linear correlation between FinBERT-extracted news sentiment and next-day stock returns across 20 major NSE equities.

- Total Data Samples Analyzed: 2576 trading days
- Overall Pearson Correlation (\$r\$): 0.0084
- Statistical Significance (\$p\$-value): 0.6713
- Overall Assessment: Weak / Moderate Correlation

2. Correlation Breakdown by NSE Stock

Symbol	Pearson \$r\$	\$p\$-value	Significant (p < 0.05)	Sample Size
TCS	0.1422	0.1167	No	123
SUNPHARMA	0.1343	0.1402	No	122
WIPRO	0.1321	0.1454	No	123
ITC	0.1067	0.2400	No	123
NTPC	0.0939	0.3017	No	123
AXISBANK	0.0917	0.3129	No	123
BAJFINANCE	0.0904	0.3202	No	123
ASIANPAINT	0.0718	0.4318	No	122
LT	0.0612	0.5029	No	122
RELIANCE	0.0606	0.5058	No	123
WAAREEENER	0.0559	0.5405	No	122
MARUTI	0.0139	0.8789	No	122
SBIN	-0.0033	0.9714	No	123
ICICIBANK	-0.0627	0.4910	No	123
HDFCBANK	-0.0703	0.4394	No	123
HCLTECH	-0.0705	0.4383	No	123
HINDUNILVR	-0.0746	0.4123	No	123
TITAN	-0.1363	0.1345	No	122
KOTAKBANK	-0.1579	0.0824	No	122
BHARTIARTL	-0.1645	0.0691	No	123
INFY	-0.1742	0.0539	No	123

3. Key Observations & Findings

1. Stocks with strong positive sentiment-return alignment display predictive signals suitable for reinforcement learning trading state representation.
2. News volume and sentiment variance serve as important indicators of price volatility on subsequent trading days.
3. Lagged features (1-day and 2-day prior sentiment) often carry stronger predictive signal than same-day sentiment due to news dissemination delays.